# ARRANGEMENT

Seating, Order and Position Puzzles

OAVS TGT | Reasoning (Verbal) | Topic 4 of the syllabus list

**Syllabus line (Reasoning, 30 Marks):** “The Verbal Reasoning topics like Series, Ranks, Direction, **Arrangement**, Coding, Decoding, Analogy and Classification/Odd Pair, Syllogism and Statement Conclusion.”

**Built from:** OAVS TGT English 2019, English 2022, Mathematics 2024, Science 2024 and Social Studies 2023 (17 May 2023, Shift 2). All 21 arrangement questions found in these papers are solved here. Every puzzle was checked by exhaustive search for a unique solution.

## What is inside


| Ch. | Title | What you get |
|---|---|---|
| **1** | PYQ Analysis | Counts, patterns, priority, index of all 21 PYQs |
| **2** | Foundation | Left/right rules, phrase-to-arithmetic table, 6-step method |
| **3** | Linear Arrangement | Rules and 4 solved PYQs (one with 33 persons) |
| **4** | Circular and Table Arrangement | Centre-facing rules; 8-person set, 7-person, square, pentagon |
| **5** | Parallel Rows | Two rows facing each other: 2 solved PYQs |
| **6** | Days and Order Puzzles | Monday to Sunday: 3 solved PYQs |
| **7** | Floors and Boxes | Vertical puzzles: 3 solved PYQs |
| **8** | Shortcuts and Traps | Speed methods and the mistakes that cost marks |
| **9** | Practice Set | 16 original questions on 10 puzzles, with answer key |
| **10** | Quick Revision | One-page cheat sheet |


**SCOPE:** Left/right *turns* and distances belong to Direction; ranks in a class to Ranks. Blood relations that appear inside a seating puzzle (for example the family around the pentagon table) are solved here as seating.

**SOURCE NOTE:** One printed question has a source problem: SST 2023 Q27 (a clause is lost and the printed version is self-contradictory), so its printed answer cannot be reproduced. It is explained where it appears in section 4.6.

CHAPTER 1

# PYQ Analysis

Arrangement is the largest verbal topic: about 4 questions per paper

| Paper | Q. numbers | Count | Puzzle types |
|---|---|---|---|
| **English 2019** | 54 | 1 | Pentagon table (family) |
| **English 2022** | 34, 43, 37, 42, 48, 49, 55 | 7 | Line of 33 (2 Qs); circle of 8 (5 Qs) |
| **Mathematics 2024** | 73, 74, 82, 84 | 4 | Two parallel rows (2); exam days (2) |
| **Science 2024** | 86, 88, 93, 95 | 4 | Floors; line of 7; circle of 7; exam days |
| **Social Studies 2023** | 6, 8, 17, 27, 28 | 5 | Line of 7; floors; square table; circle of 7; boxes |
| **Total** | | **21** | **Every paper has at least one; 4 or more in four of five papers** |

| Puzzle type | PYQs | Priority |
|---|---|---|
| Circular / polygon table (facing the centre) | 9 | ★★★ Must |
| Linear row (facing north or south) | 4 | ★★★ Must |
| Days of the week (Monday to Sunday) | 3 | ★★ High |
| Two parallel rows facing each other | 2 | ★★ High |
| Floors of a building | 2 | ★★ High |
| Boxes one above the other | 1 | ★ Medium |
| *Expected, not yet asked:* months / dates, faces-outward circle, rectangular table, mixed attributes (colour, profession) | 0 | ★ Expected |

**PYQ INSIGHT:** **One puzzle, many questions:** English 2022 gives one circle of 8 that feeds **five** questions and one line of 33 that feeds two. Solve the arrangement once and the rest take 10 seconds each.

**PYQ INSIGHT:** Exam-day puzzles appeared in **both 2024 papers** (3 questions). All have 7 persons Monday to Sunday, with clues of the form “only k people between”, “immediately before / after” and “only k people after”.

**PYQ INSIGHT:** Almost all clues are **exact** (“second to the left”, “only three between”, “immediate right”). No PYQ needed trial-and-error beyond a two-way branch.

## Index of the 21 PYQs

| Paper / Q | Puzzle and question in short | Ans | See |
|---|---|---|---|
| Eng 2019 Q54 | Pentagon table: who is immediate left of the daughter | Sridhar | 4.5 |
| Eng 2022 Q34 | Line of 33: persons between H (extreme left) and C | 12 | 3.4 |
| Eng 2022 Q43 | Same line: total persons | 33 | 3.4 |
| Eng 2022 Q37 | Circle of 8: which statement is NOT correct | C is immediate right of G | 4.2 |
| Eng 2022 Q42 | Circle of 8: immediate left of G | A | 4.2 |
| Eng 2022 Q48 | Circle of 8: 4th to the right of B | H | 4.2 |
| Eng 2022 Q49 | Circle of 8: 3rd to the left of C | G | 4.2 |
| Eng 2022 Q55 | Circle of 8: immediate right of C | D | 4.2 |
| Maths 2024 Q73 | Two rows of 4: which pair faces each other | H, Q | 5.1 |
| Maths 2024 Q74 | Two rows of 4: who faces C | Q | 5.2 |
| Maths 2024 Q82 | Exam days: persons between T and U | Three | 6.1 |
| Maths 2024 Q84 | Exam days: who has the exam on Sunday | S | 6.2 |

| Paper / Q | Puzzle and question in short | Ans | See |
|---|---|---|---|
| Science 2024 Q95 | Exam days: persons between Q and T | 1 | 6.3 |
| Science 2024 Q86 | Six floors: floor of Q | Floor 5 | 7.1 |
| Science 2024 Q88 | Line of 7 facing north: persons to the right of E | One | 3.2 |
| Science 2024 Q93 | Circle of 7: persons between A and B from the right of A | 2 | 4.3 |
| SST 2023 Q6 | Line of 7: immediate right of Rachit | Yashika | 3.3 |
| SST 2023 Q8 | Six floors: floor of B | Floor 1 | 7.2 |
| SST 2023 Q17 | Square table: immediate left of Anand | Raj | 4.4 |
| SST 2023 Q27 | Circle of 7: third to the left of K (clause lost) | O (key) | 4.6 |
| SST 2023 Q28 | Six boxes: object in the second box from the top | Chalks | 7.3 |

CHAPTER 2

# Foundation

Rules that make every arrangement puzzle mechanical

## 2.1 Left and right

| Arrangement | Facing | Left of a person | Right of a person |
|---|---|---|---|
| **Linear row** | North | West (towards the left of the page) | East |
| **Linear row** | South | East (towards the right of the page) | West |
| **Circle / table** | Centre | **Clockwise** neighbour (+1) | **Anticlockwise** neighbour (−1) |
| **Circle / table** | Outside | Anticlockwise neighbour (−1) | Clockwise neighbour (+1) |
| **Two rows (row 1 faces South, row 2 faces North)** | — | Row 1: East. Row 2: West. | Row 1: West. Row 2: East. |

**SHORTCUT: Always convert to numbers.** For a row, number the positions from the persons' own left end (1 = left end). 'Left' then means a smaller number and 'right' a larger number, whichever way they face. For a circle facing the centre, number clockwise; 'left' = +1, 'right' = −1.

## 2.2 Phrase to arithmetic

| Phrase | Row (position from left end) | Circle facing centre (clockwise +) |
|---|---|---|
| **Immediate left of X** | X − 1 | X + 1 |
| **Immediate right of X** | X + 1 | X − 1 |
| **n-th to the left of X** | X − n | X + n |
| **n-th to the right of X** | X + n | X − n |
| **Only k persons between A and B** | difference = k + 1 | k + 1 steps along one side (two possible sides) |
| **n-th from the left end / right end** | n / N − n + 1 | — |
| **Opposite X (N even)** | — | X + N ÷ 2 |
| **Neighbours of X** | X − 1 and X + 1 | X − 1 and X + 1 |
| **Between A and B 'counted from the right of A'** | — | start at A, move anticlockwise (−1), count persons before B |

## 2.3 The six-step method

| Step | Do this |
|---|---|
| **1** | Note the number of persons / slots and which way they face. Draw an empty frame (boxes in a row, circle with N dots, or a 2 × 4 grid). |
| **2** | Pick a **reference**: any seat for a circle (call it 0), the left end for a row. |
| **3** | Place **definite** clues first: exact positions, 'extreme end', 'opposite', 'immediate', 'only k between' with an end. |
| **4** | Join clues that share a name into a **chain** (X is left of Y, Y is left of Z). |
| **5** | If a clue gives two options, **branch**. Test each against every other clue; the failed branch dies. |
| **6** | Re-read all clues against the final arrangement, then answer. Never answer before verifying. |

**Clue strength (use strongest first):** exact positions, 'extreme end', 'opposite', 'immediate' → 'second to the left', 'only k between' → 'somewhere to the left', 'not at the end', 'not a neighbour'.

CHAPTER 3

# Linear Arrangement

People in one row: 4 PYQs

## 3.1 Method

- Draw boxes numbered from the left end. Place any person with an exact position (‘eighth from the left’, ‘extreme right’).
- ‘Only k persons between X and Y’ fixes Y at X ± (k + 1); test both and drop the one that breaks another clue.

## 3.2 Science 2024 Q88: seven people, facing north

A, B, C, D, E, F and G sit in a straight line facing north. Only two people sit to the left of G. Only three people sit between G and A. Only one person sits between A and B. D sits to the immediate left of F. C is not an immediate neighbour of A. How many people sit to the right of E?

| Seat | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|
| **Person** | D | F | G | C | B | E | A |

**1.** Two people to the left of G: G is seat 3.
**2.** Three between G and A: A is seat 7 (seat −1 does not exist). One between A and B: B is seat 5.
**3.** D is immediate left of F (F = D + 1). Free seats are 1, 2, 4, 6 and the only adjacent pair is 1, 2: D = 1, F = 2.
**4.** C and E take seats 4 and 6. C is not next to A (seat 7), so C ≠ 6: C = 4, E = 6.
**5.** Only A (seat 7) is to the right of E.
**Answer: One person (Option 4)**

## 3.3 SST 2023 Q6: seven people, facing north

Manjot, Nivya, Pranjal, Rachit, Saif, Shanaya and Yashika sit in a line facing north. Saif is to the immediate left of Rachit. Yashika is third to the left of Shanaya. Nivya is second to the right of Manjot, who is to the immediate right of Pranjal. Pranjal is third to the right of Saif. Who is to the immediate right of Rachit?

| Seat | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|
| **Person** | Saif | Rachit | Yashika | Pranjal | Manjot | Shanaya | Nivya |

**1.** Let Saif be seat s. Then Rachit = s + 1, Pranjal = s + 3, Manjot = s + 4, Nivya = s + 6 (a chain).
**2.** Nivya’s seat s + 6 cannot exceed 7, so s = 1: Saif 1, Rachit 2, Pranjal 4, Manjot 5, Nivya 7.
**3.** Seats 3 and 6 remain. Yashika is three seats left of Shanaya: Yashika 3, Shanaya 6.
**4.** The immediate right of Rachit (seat 2) is seat 3.
**Answer: Yashika (Option 1)**

## 3.4 English 2022 Q34 and Q43: a line of 33

A certain number of people are sitting in a straight line, facing north. A is sitting eighth from the left end of the line. Only five people are sitting between A and C. D is sitting second to the left of C. E is sitting fifth to the right of D. E is seventeenth from the right end of the line. **Q34:** If H is sitting at the extreme left end, then how many people are sitting between H and C? **Q43:** Based on the same seating arrangement, how many persons are sitting in the line in total?

| Person | H | A | D | C | E |
|---|---|---|---|---|---|
| **Seat from left** | 1 | 8 | 12 | 14 | 17 |

**1.** A = 8. Five between A and C: C = 14 or 2. D is second to the left of C: D = C − 2, so C = 2 would give D = 0, impossible. C = 14, D = 12.
**2.** E is fifth to the right of D: E = 17. E is 17th from the right end, so the total = 17 + 17 − 1 = **33** (Q43).
**3.** H is at seat 1 and C at seat 14: persons between = 14 − 1 − 1 = **12** (Q34).
**Answer: Q43 = 33; Q34 = 12**

CHAPTER 4

# Circular and Table Arrangement

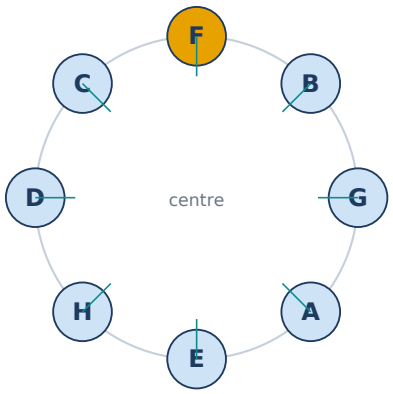
People facing the centre: 9 PYQs

## 4.1 Method

- Give each seat a number clockwise, 0 to N − 1. Facing the centre: **left = +1, right = −1** (see 2.1).
- Set one person at 0. Translate each clue into ‘seat = other seat ± n’ using the table in 2.2.
- Ask the final question by number: n-th to the left of X is seat X + n (mod N).

## 4.2 English 2022: circle of 8 (Q37, Q42, Q48, Q49, Q55)

A, B, C, D, E, F, G and H sit around a circle facing the centre. C is to the immediate right of F, who is 6th to the right of D. G is 2nd to the left of F, and E sits in front of F. B and H sit just opposite each other. H is 2nd to the right of C.




Clockwise order, all face the centre

1. Put F at 0. C is immediate right of F: C = −1 = 7.
2. F is 6th to the right of D: F = D − 6, so D = F + 6 = 6.
3. G is 2nd to the left of F: G = 2. E is in front of F: E = 4.
4. H is 2nd to the right of C: H = 7 − 2 = 5. B is opposite H: B = 5 + 4 = 9 − 8 = 1.
5. A is the only one left: seat 3.

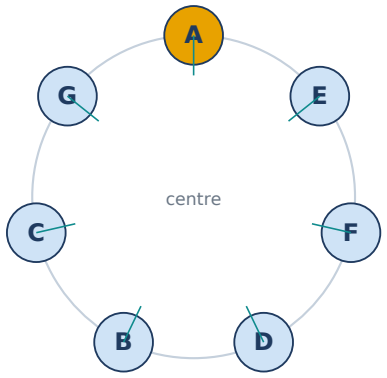
| Seat | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| **Person** | F | B | G | A | E | H | D | C |

| Q | Question | Working | Ans |
|---|---|---|---|
| Q37 | Which statement is NOT correct? | A: C is immediate right of G? G = 2 so its immediate right is seat 1 = B, not C → **not correct**. B: A (3) and C (7) are opposite ✓. C: G (2) and B (1) adjacent ✓. D: D (6) is 2nd to the left of E (4 + 2) ✓. | A |
| Q42 | Immediate left of G | G + 1 = seat 3 | A |
| Q48 | 4th to the right of B | B − 4 = 1 − 4 = −3 = seat 5 | H |
| Q49 | 3rd to the left of C | C + 3 = 7 + 3 = 10 = seat 2 | G |
| Q55 | Immediate right of C | C − 1 = seat 6 | D |

**SHORTCUT: Q37 is a ‘NOT correct’ question:** test each statement and stop at the one that fails. Do not read all four if the first fails.

## 4.3 Science 2024 Q93: circle of 7

A, B, C, D, E, F and G sit around a circular table facing the centre. Only two people sit between F and G when counted from the right of F. Only three people sit between D and A when counted from the right of A. G sits to the immediate right of A. B sits to the immediate right of C. How many people sit between A and B when counted from the right of A?



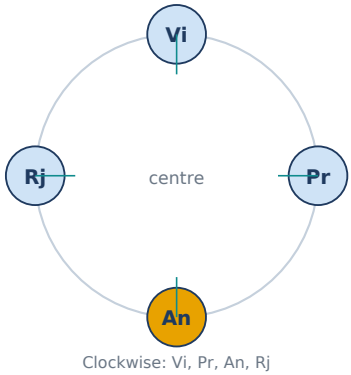
Clockwise order, all face the centre

**1.** A = 0. G is immediate right of A: G = −1 = 6.
**2.** From F, counting anticlockwise (right), two people then G: F = 6 + 3 = 9 = 2 (mod 7).
**3.** From A, counting anticlockwise, three people then D: D = 0 − 4 = −4 = 3.
**4.** B is immediate right of C: seats 4, 5 are free with E in 1, so B = 4, C = 5, E = 1.
**5.** Counting anticlockwise from A: G (6), C (5), then B (4): **2** people between.

**Answer: 2 (Option 1)**

## 4.4 SST 2023 Q17: square table

Vishal, Anand, Raj and Prateek sit around a square table facing the centre, at the middle of each side. Prateek is to the immediate left of Vishal. Prateek is not to the immediate right of Raj. Who is to the immediate left of Anand?



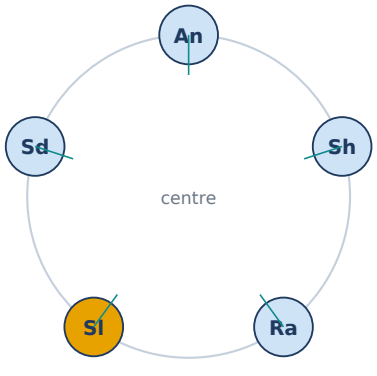
Clockwise: Vi, Pr, An, Rj

**1.** Prateek is immediate left of Vishal: going clockwise, Vishal is followed by Prateek.
**2.** Prateek is not immediate right of Raj: Raj is not the person clockwise after Prateek. So Anand follows Prateek, and Raj follows Anand.
**3.** Order clockwise: Vishal, Prateek, Anand, Raj. The immediate left of Anand is the next clockwise = Raj.

**Answer: Raj (Option 2)**

## 4.5 English 2019 Q54: pentagon table

Ramarao has two sons and a daughter. All the family sit at the corners of a pentagonal table. Ramarao sits to the left of his wife Shyamala. His elder son Anand sits to the right of his mother. Shyamala’s daughter Srilekha sits immediate left of Ramarao. Younger son Sridhar sits between Anand and Srilekha. Who is immediate left of Ramarao’s daughter?



Clockwise order

**1.** Ramarao is immediate left of Shyamala: clockwise, Shyamala then Ramarao.
**2.** Anand is immediate right of Shyamala (anticlockwise side): Anand, Shyamala, Ramarao.
**3.** Srilekha is immediate left of Ramarao: Anand, Shyamala, Ramarao, Srilekha.
**4.** Sridhar fills the last seat, between Srilekha and Anand ✓.
**5.** The immediate left of Srilekha (next clockwise) is Sridhar.

**Answer: Sridhar (Option B)**

## 4.6 SST 2023 Q27: circle of 7 (source problem)

**SOURCE NOTE:** The printed question reads ‘L is sitting second to the right of [someone], who is sitting second to the left of O’. The name was lost, and as printed the two clues put L on O’s own seat, which is impossible for any name. Using the solid clues (N is 4th to the left of M; only K sits between N and R) several arrangements remain, so the printed answer (O) cannot be reproduced. Treat it as unreliable and do not study it further.

CHAPTER 5

# Parallel Rows

Two rows facing each other: 2 PYQs

## 5.1 Method and Maths 2024 Q73

- Draw a 2 × N grid with columns 1 to N from the west (left of the page) to the east.
- **A faces B** means A and B are in the same column. Row 1 faces south (left = east, right = west); row 2 faces north (left = west, right = east).
- Fix the clue that tells you columns (‘extreme end’, ‘faces’), then use neighbour clues.

Row 1 (E, F, G, H) faces south; row 2 (Q, R, S, T) faces north. T sits to the immediate right of R. Only two people sit between G and E. R faces G. The one who faces H sits to the immediate left of S. Q does not sit at an extreme end. Which pair faces each other?

| | Col 1 | Col 2 | Col 3 | Col 4 |
|---|---|---|---|---|
| **Row 1 (faces S)** | G | F | H | E |
| **Row 2 (faces N)** | R | T | Q | S |

**1.** Only two between G and E: they sit at the two ends (columns 1 and 4), in some order.
**2.** R faces G, so R is also in column 1 or 4. T is immediate right (east) of R, so R cannot be in column 4: R = 1, T = 2, and G = column 1, E = column 4.
**3.** Q is not at an end, so Q = column 3 and S = column 4.
**4.** The person facing H sits immediate left (west) of S, so that person is in column 3 (Q); H = column 3 and F = column 2. H faces Q.
**Answer: H, Q (Option D)**

## 5.2 Maths 2024 Q74

Row 1 (A, B, C, D) faces south; row 2 (P, Q, R, S) faces north. D sits third to the right of B. The one facing P is an immediate neighbour of B. The one facing C is an immediate neighbour of P. The one facing D is an immediate neighbour of Q. R sits somewhere to the left of S. Who faces C?

| | Col 1 | Col 2 | Col 3 | Col 4 |
|---|---|---|---|---|
| **Row 1 (faces S)** | D | C | A | B |
| **Row 2 (faces N)** | R | Q | P | S |

**1.** Row 1 faces south, so right = west. D is third to the right of B: B = column 4, D = column 1.
**2.** The one facing P is next to B (column 3), so P = column 3 and A or C is there. The one facing C is a neighbour of P, so C = column 2 (not 3), A = column 3.
**3.** The one facing D (column 1) is a neighbour of Q: Q = column 2.
**4.** R is left (west) of S: R = column 1, S = column 4.
**5.** C (column 2) faces Q.
**Answer: Q (Option D)**

**TRAP:** The word **faces** pairs people in the **same column**. Left and right in row 2 are the opposite way round from row 1. Write ‘Row 1: left = east’ on the page.

CHAPTER 6

# Days and Order Puzzles

Seven people, Monday to Sunday: 3 PYQs

## 6.1 Method

- Number the days Mon = 1 ... Sun = 7. 'Only k people have exams after P' puts P at day 7 − k.
- 'Only k between X and Y': difference = k + 1. 'Immediately before / after': difference = 1, in that order.
- Fix the person with the strongest clue first (a given day, or an end), then everything else.

### Maths 2024 Q82

P, Q, R, S, T, U and V have exams on different days Monday to Sunday. Only two people have exams after P. Only three people have exams between V and P. Q has the exam before T but after P. U has the exam before R but after S. How many people have exams between T and U?

| Day | Mon | Tue | Wed | Thu | Fri | Sat | Sun |
|---|---|---|---|---|---|---|---|
| **Person** | V | S | U | R | P | Q | T |

**1.** Two after P: P = Friday (5). Three between V and P: V = Monday (1).
**2.** Q is after P and before T: Q = Saturday, T = Sunday.
**3.** S, U, R remain for Tue, Wed, Thu with S < U < R: S = Tue, U = Wed, R = Thu.
**4.** Between T (Sun) and U (Wed): Thu, Fri, Sat = 3.
**Answer: Three (Option A)**

## 6.2 Maths 2024 Q84

P, Q, R, S, T, U and V have exams on different days Monday to Sunday. U has the exam on Wednesday. Exactly 3 people have exams between U and S. Q has the exam immediately after R and T immediately before R. Only 3 people have exams between P and R. Who has the exam on Sunday?

| Day | Mon | Tue | Wed | Thu | Fri | Sat | Sun |
|---|---|---|---|---|---|---|---|
| **Person** | P | V | U | T | R | Q | S |

**1.** U = Wed (3). Three between U and S: S = Sunday (7) (day −1 does not exist).
**2.** T, R, Q are consecutive in that order. The free days are 1, 2, 4, 5, 6: the only three in a row are 4, 5, 6: T = Thu, R = Fri, Q = Sat.
**3.** Three between P and R (Fri): P = Monday. V = Tuesday.
**Answer: S (Option A)**

## 6.3 Science 2024 Q95

P, Q, R, S, T, U and V have exams on different days Monday to Sunday. V has the exam on Tuesday. Exactly 2 people have exams between V and P. Q has the exam immediately before U and S immediately after R. Only 1 person has an exam between U and R. How many people have exams between Q and T?

| Day | Mon | Tue | Wed | Thu | Fri | Sat | Sun |
|---|---|---|---|---|---|---|---|
| **Person** | T | V | Q | U | P | R | S |

**1.** V = Tue (2). Two between V and P: P = Friday (5).
**2.** Pairs (Q, U) and (R, S) are consecutive; the free days are Mon, Wed, Thu, Sat, Sun. The pairs take (Wed, Thu) and (Sat, Sun).
**3.** Only one person between U and R: Q, U = Wed, Thu and R, S = Sat, Sun gives Thu, Fri, Sat (P between) ✓. The other way round leaves 3 between.
**4.** T = Monday. Between Q (Wed) and T (Mon): V only = 1.
**Answer: 1 (Option 2)**

> **SHORTCUT:** When two **consecutive pairs** (Q, U) and (R, S) must fit the free days, list the free days first. Only certain pairs of consecutive days exist, and the 'only one between' clue picks the layout.

CHAPTER 7

# Floors and Boxes

Vertical puzzles: 3 PYQs

- **Floors:** number from the ground floor upwards (floor 1 lowest, 6 topmost). ‘Immediately above’ = +1.
- **Boxes:** number from the top (box 1 topmost) unless the question says otherwise. ‘Immediately above’ = −1 in that numbering.
- ‘Even / odd floor’, ‘uppermost’, ‘not floor 1’ are strong clues: use them to shrink options first.

## 7.1 Science 2024 Q86: six floors

P, Q, R, S, T and U live on six different floors (1 lowest, 6 topmost). U lives on an even-numbered floor but not floor 2. Only one person lives between U and T. P lives on the floor immediately above T. R lives on the uppermost floor. Q lives on an odd-numbered floor but not floor 1. On which floor does Q live?

| Floor | 6 | 5 | 4 | 3 | 2 | 1 |
|---|---|---|---|---|---|---|
| **Person** | R | Q | U | P | T | S |

**1.** R = 6 (uppermost). U is even and not 2 and not 6: U = 4.
**2.** Only one person between U and T: T = 2 (T = 6 is taken). P is immediately above T: P = 3.
**3.** Q is odd and not 1: floors 1 and 5 are left, so Q = 5 and S = 1.
**Answer: Floor 5 (Option 1)**

## 7.2 SST 2023 Q8: six floors

A, B, C, D, E and F live on different floors (1 lowest, 6 topmost). F lives below C. D lives just above F. E lives on the topmost floor. Only one person lives between C and E. D lives below A. F does not live on floor 1. On which floor does B live?

| Floor | 6 | 5 | 4 | 3 | 2 | 1 |
|---|---|---|---|---|---|---|
| **Person** | E | A | C | D | F | B |

**1.** E = 6. One person between C and E: C = 4.
**2.** F is below C and not floor 1: F = 2 or 3. D is just above F and below A. If F = 3 then D = 4 = C, not possible: F = 2, D = 3.
**3.** A is above D and not on 4 or 6: A = 5. B takes the last floor, 1.
**Answer: Floor 1 (Option 4)**

## 7.3 SST 2023 Q28: six boxes

Six boxes hold pens, pencils, erasers, sharpeners, chalks and markers, one each, kept one above the other. Only two boxes between erasers and pens. Only three between pencils and chalks. Erasers are immediately above chalks. Erasers are above pens. Sharpeners are above markers. Which object is in the second box from the top?

| Box (from top) | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| **Object** | Erasers | Chalks | Sharpeners | Pens | Markers | Pencils |

**1.** Three between pencils and chalks: (box 1, box 5) or (box 2, box 6) in either order.
**2.** Chalks are immediately below erasers, and pens are 3 below erasers. Erasers can be box 1 (pens 4) or box 4 (pens 7, impossible) or box 2 (chalks 3 would break the pencils rule).
**3.** Try erasers = 1: chalks = 2, pencils = 6 (three between 2 and 6 ✓), pens = 4.
**4.** Sharpeners above markers in the remaining boxes 3 and 5: sharpeners = 3, markers = 5.
**Answer: Chalks (Option 3)**

CHAPTER 8

# Shortcuts and Traps

Marks saved and marks lost

- **Number everything.** Seat numbers beat pictures: ‘second to the left’ becomes +2.
- **Use exact clues first**; keep ‘not’ and ‘somewhere’ clues for the end, where they eliminate branches.
- **Chains:** if each clue refers to the previous person, write all seats in terms of one unknown and let the length of the row fix it (SST 2023 Q6).
- **Opposite** in a circle of N is N ÷ 2 away. Use it to place two people at once.
- **Branch early:** when ‘only k between’ gives two options, test both against the next clue before going further.
- **One puzzle, many questions:** solve the full arrangement once; then read each answer off the table (English 2022 had five).
- **Mod arithmetic:** in a circle of N, add or subtract N when you go below 0 or above N − 1.
- **Days:** ‘Only k after P’ means P = 7 − k. ‘Immediately before’ pairs are consecutive.

| Trap | How to avoid it |
|---|---|
| **Left / right the wrong way round** | Write the rule on the page first: circle facing centre, left = clockwise. For two rows: row 1 left = east, row 2 left = west. |
| **‘Between’ counted on the wrong side of a circle** | The question says ‘counted from the right of A’. Start at A and go anticlockwise; do not count the shorter arc by habit. |
| **‘Only k persons between’ taken as difference k** | The positions differ by k + 1. |
| **Mixing ‘from the left end’ and ‘left of’** | ‘Eighth from the left end’ is a position; ‘to the left of X’ is a direction. Convert both to numbers. |
| **Not checking the second branch** | If both options survive the clue, keep both until the next clue kills one. |
| **Answering ‘NOT correct’ questions by checking only one option** | Test each statement until you find the false one. |
| **Spending over 2 minutes on one clue** | Skip a clue that will not fit, finish with the others, then re-check it. |

**SHORTCUT: Time plan:** a 7-person puzzle with 2 questions should take about 3 minutes. If a puzzle has 4 or 5 questions (like English 2022), it is worth 6 to 8 minutes: solve it first.

CHAPTER 9

# Practice Set

16 original questions on 10 puzzles; each puzzle has a unique solution

**Puzzle 1 (Q1–Q3).** Six friends A, B, C, D, E and F sit in a row facing north. A sits third from the left end. B sits to the immediate right of A. D sits at the extreme right end. F sits to the immediate left of D. Only one person sits between E and A.

1. Who sits second from the left end?
2. How many people sit between C and F?
3. Who sits to the immediate right of B?

**Puzzle 2 (Q4–Q6).** P, Q, R, S, T, U, V and W sit around a circle facing the centre. S sits third to the left of P. T sits opposite P. R sits second to the right of T. Q sits between P and R. V sits to the immediate left of U. W sits to the immediate right of P.

4. Who sits second to the left of V?
5. Who sits opposite R?
6. How many people sit between U and Q when counted from the right of U?

**Puzzle 3 (Q7–Q8).** Row 1 has A, B, C, D facing south. Row 2 has P, Q, R, S facing north. Each person faces someone in the other row. D sits to the immediate left of B. A sits second to the right of B. S sits at the extreme right end of row 2. P faces C. The one facing A is an immediate neighbour of Q.

7. Who faces D?
8. Who sits to the immediate left of R?

**Puzzle 4 (Q9–Q10).** P, Q, R, S, T, U and V have exams on different days Monday to Sunday. Only two people have exams after Q. P has the exam on Tuesday. Only one person has an exam between P and V. S has the exam immediately after Q. T has the exam before P. R has the exam before V.

9. Who has the exam on Sunday?
10. How many people have exams between T and R?

**Puzzle 5 (Q11–Q12).** A, B, C, D, E and F live on six different floors (1 lowest, 6 topmost). D lives on the topmost floor. A lives on an odd-numbered floor but not floor 1. Only one person lives between A and B. C lives immediately above E. F lives immediately above A.

11. Who lives on floor 2?
12. On which floor does B live?

**Puzzle 6 (Q13–Q14).** Six boxes hold tea, salt, rice, coffee, wheat and sugar, one each, kept one above the other. Tea is in the top box. Salt is immediately above rice. Only two boxes are between rice and sugar. Coffee is somewhere above wheat. Wheat is immediately above sugar.

13. What is in the third box from the top?
14. What is kept immediately below coffee?

**Puzzle 7 (Q15).** A, B, C and D sit around a square table facing the centre. A sits opposite C. B is to the immediate left of A.

15. Who is to the immediate right of A?

**Puzzle 8 (Q16).** P, Q, R, S and T sit at the corners of a pentagonal table facing the centre. Q is to the immediate right of P. R is to the immediate left of P. S sits between Q and T.

16. Who is to the immediate left of R?

## Answer key with the arrangement

| Puzzle | Arrangement (solved) | Answers |
|---|---|---|
| **1 (row, left → right)** | E, C, A, B, F, D | Q1: C Q2: 2 Q3: F |
| **2 (circle, clockwise from P)** | P, Q, R, S, T, U, V, W | Q4: P Q5: V Q6: 3 (T, S, R) |
| **3 (columns 1 → 4)** | Row 1: C, A, D, B  Row 2: P, R, Q, S | Q7: Q Q8: P |
| **4 (Mon → Sun)** | T, P, R, V, Q, S, U | Q9: U Q10: 1 (P) |
| **5 (floor 1 → 6)** | E, C, A, F, B, D | Q11: C Q12: floor 5 |
| **6 (box 1 → 6)** | Tea, salt, rice, coffee, wheat, sugar | Q13: rice Q14: wheat |
| **7 (circle, clockwise from A)** | A, B, C, D | Q15: D |
| **8 (circle, clockwise from P)** | P, R, T, S, Q | Q16: T |

### Key steps

- **P1:** A = 3, B = 4, D = 6, F = 5. E is two seats from A, so E = 1 (seat 5 is taken). C = 2.
- **P2:** P = 0, S = 3, T = 4, R = T − 2 = 2, Q = 1, W = P − 1 = 7. U, V in 5, 6 with V = U + 1: U = 5, V = 6.
- **P3:** B = D + 1 (row 1 faces south, left = east); A = B − 2. So D = 3, B = 4, A = 2, C = 1. S = 4; P faces C so P = 1; R in 2 (facing A) is the neighbour of Q (3).
- **P4:** Q = 5 (two after). P = 2. V = 4 (one between). S = 6. T = 1. R before V: R = 3. U = 7.
- **P5:** D = 6. A = 3 or 5; F = A + 1 forces A = 3, F = 4. B = 5 (B = 1 would leave 2 and 5 for C, E, not consecutive). E = 1, C = 2.
- **P6:** Tea = 1. Rice = 3, Salt = 2, Sugar = 6, Wheat = 5, Coffee = 4.
- **P7:** C = 2 (opposite A = 0), B = 1 (A + 1), D = 3. The immediate right of A is seat −1 = 3 = D.
- **P8:** Clockwise: Q, P, R, then T and S (S is between Q and T, so S is next to Q): Q, P, R, T, S. Immediate left of R = T.

CHAPTER 10

# Quick Revision

One page for the last day

| Type | Rule to apply | PYQ check |
|---|---|---|
| **Circle facing centre** | Left = +1 (clockwise), right = −1. Opposite = N ÷ 2. | Eng 2022: F, B, G, A, E, H, D, C (8 seats) |
| **Row** | Number from the left end; left = smaller. Total = a + b − 1 when positions from both ends are known. | Eng 2022 Q43: E = 17, 17th from right → 33 |
| **Chain in a row** | Write all seats as s + k; the row length fixes s. | SST 2023 Q6: Saif = 1 |
| **Two rows** | Same column = faces. Row 1 (S): left = east; Row 2 (N): left = west. | Maths 2024 Q73: H faces Q |
| **Days** | Mon = 1 … Sun = 7; ‘only k after P’ → P = 7 − k. | Maths 2024 Q82: V, S, U, R, P, Q, T |
| **Floors** | 1 = lowest; ‘immediately above’ = +1. | Science 2024 Q86: Q on floor 5 |
| **Boxes** | 1 = top; ‘immediately above’ = −1. | SST 2023 Q28: chalks in box 2 |
| **Square / pentagon table** | Same as a circle of 4 / 5. | Eng 2019 Q54: Sridhar |

| Phrase | Row | Circle (centre) |
|---|---|---|
| **Immediate left of X** | X − 1 | X + 1 |
| **n-th to the right of X** | X + n | X − n |
| **Only k between A and B** | \|A − B\| = k + 1 | k + 1 steps |
| **n-th from the right end** | N − n + 1 | — |

**ROUTINE: Exam-day routine:** count persons → number the seats → place exact clues → chain clues → branch only when forced → verify every clue → answer. Solve the biggest multi-question puzzle first.